ENGINEERING
McMaster-Mohawk
Bachelor of Technology
Partnership

McMaster University

Course Outline

1. COURSE INFORMATION

Session Offered	Fall 2026	
Course Name	Capstone Design Project-II	
Course Code	PROC TECH 4TR3	
Date(s) and Time(s) of lectures	C03 – Thursday: 4:30 PM – 8:20 PM in ETB/B111 C04 – Tuesday: 10:30 PM – 2:20 PM in ETB/B111	
Program Name	Automation Systems Engineering Technology Co-op	
Calendar Description	This course is a continuation of PROCTECH 4TR1 and it requires students to conduct further research, modify/refine project design, develop and implement the independent project proposal submitted as a part of the Capstone Design Project I course. The project will be documented as a technical report and presented in a seminar.	
Instructor(s)	Kostas Apostolou	E-Mail: apostol@mcmaster.ca Office Hours & Location: ETB/213

2. COURSE SPECIFICS

Course Description

The main objective of this course is to teach students the concept of design optimization and implementation. The course is a continuation of Capstone Design Project I and it requires the application of the knowledge gained in earlier courses to the design and implementation process of the students' project. Skills applied include:

- System definition, analysis, synthesis, integration.
- Communication
- Organizational and tea-work

Furthermore the course stresses the importance of other influences on design and implementation such as economics, reliability, performance, safety, ethics, social impact, and sustainability.

Instruction Type

Code	Type	Hours per term
C	Classroom instruction	
L	Laboratory, workshop or fieldwork	36
T	Tutorial	12
DE	Distance education	
	Total Hours	48

Resources

ISBN	Textbook Title & Edition	Author & Publisher
ISBN:		

Other Supplies	Source

Prerequisite(s)	PROCTECH 4TR1, 4IT3 or SFWRTECH 4WP3; PROCTECH 4IC3 or SMRTTECH 4ID3 or SFWRTECH 3IT3, ENGTECH 4EE0 and registration in level IV of Automation Systems Engineering Technology Co-op program Cannot be taken concurrently with any co-op work term course.

Corequisite(s)	N/A
Antirequisite(s)	N/A
Course Specific Policies	1. **All projects must be completed by December 1 or 3, depending on your class section.** 2. Course attendance is mandatory. Students are expected to attend the full four-hour course section and spend that time working on their project and discussing project issues with their instructor and group peers. Failure to attend course sections, without submitted MSAF, will affect the final student's course grade. 3. While students work in groups to complete their project, it is expected that they will work independently on different aspects and the final project will be formed by the integration, by a group member, of these different parts. 4. Students will be meeting face-to-face. 5. Each student will **individually** keep a log of work performed each week outside and inside the lab/class. The logbook will have the following information: project progress as compared to the plan, what **individual** tasks were accomplished and what was learned. The logbook will also contain any suggestions that were made and any action taken on them. Logbook entries must be uploaded on Avenue each week at the end of class after discussion with the instructor. For students keeping a hard copy of the logbook, the book entries must be signed by the instructor before leaving class every week. A mark of 0% will be assigned for the logbook component of the assessment if a logbook is not submitted in a timely manner. 6. It is expected that each student will also work on their project outside the assigned classroom/lab time. These activities will be recorded in the logbook as described above. 7. Each group shall develop a plan for executing their project. This plan must be submitted by the end of the second week of the term and must clearly identify the scope of work and responsibilities of each group member. Group members will be judged against this plan for marking of the final project. 8. Each group will submit a Mid Term Report. The Mid Term report is required by the end of week 6 in the semester. The list of items that need ordering, if any, should be submitted ASAP. The report should have the structure that would form the basis of the final report. Any change in the project scope as outlined in the PROCTECH 4TR1 course should be clearly identified. 9. Each group will submit a Final Report* at the end of the semester. The evaluation of this report will be based on: a) project plan, its implementation and accomplishments, technical content, depth and comprehension, originality, working demonstration, and problem solving skills; b) background, report structure & format, & written communication skills; and c) literature references (internet, text & reference books, reports, & original journals).

	10. Each group will present their project to the class. The actual date for the presentation shall be communicated during the semester. 11. Without use of logbook the final report will not be accepted. 12. Each group shall make a short video of the project implementation. The video shall demonstrate the operation of the implemented device. It shall not be longer than 3 minutes and it shall be played as part of the final presentation. 13. Late reports will result in 5% loss of marks per day. Submissions later than 7 days will be given a mark of zero. 14. Students may use generative AI to brainstorm ideas, explore concepts, and refine drafts. Students should not use generative AI to submit work without critically evaluating, adapting, and taking responsibility for the final product. Where generative AI is used, students are expected to acknowledge / cite its use as per the Students section of McMaster's guidelines on generative AI 15. The departmental Safety Policy must be followed by the students while working in the lab. *The submitted reports after grading will become the property of the department and will not be returned to the students. The student will be able to view them after grading.
Departmental Policies	Students must maintain a GPA of 3.5/12 to continue in the program. In order to achieve the required learning objectives, on average, B.Tech. students can expect to do at least 3 hours of "out-of-class" work for every scheduled hour in class. "Out-of-class" work includes reading, research, assignments and preparation for tests and examinations. Where group work is indicated in the course outline, such collaborative work is mandatory. The use of cell phones, iPods, laptops and other personal electronic devices are prohibited from the classroom during the class time, unless the instructor makes an explicit exception. Announcements made in class or placed on Avenue are considered to have been communicated to all students including those individuals that are not in class. Instructor has the right to submit work to software to identify plagiarism.

3. SUB TOPIC(S)		
Week 1	Project Work & Consultation	
Week 2	Project Work & Consultation **Project Plan due**	
Week 3	Project Work & Consultation	
Week 4	Project Work & Consultation	
Week 5	Project Work & Consultation	
Mid-term recess (reading week)		
Week 6	Project Work & Consultation	

	Midterm Report due	
Week 7	Project Work & Consultation	
Week 8	Project Work & Consultation	
Week 9	Project Work & Consultation	
Week 10	Project Work & Consultation	
Week 11	Project Work & Consultation	
Week 12	Project Work & Consultation ***Last day of the class is December 1 or 3, depending on your section***	

Classes end: Thursday, December 10
Final examination period: Friday, December 11 to Wednesday, December 23
All examinations MUST be written during the scheduled examination period.

Note that this structure represents a plan and is subject to adjustment term by term.
The instructor and the University reserve the right to modify elements of the course during the term. The University may change the dates and deadlines for any or all courses in extreme circumstances. If either type of modification becomes necessary, reasonable notice and communication with the students will be given with explanation and the opportunity to comment on changes.

4. ASSESSMENT OF LEARNING *including dates*	**Weight**
Project Plan (due Sep. 18)	5%
Mid Term Report (due Oct. 23)	15%
Logbook, Attendance, Participation	15%
Project Presentation	10%
Demonstration	20%
Final Report (due Dec 10)	35%
TOTAL	**100%**

Percentage grades will be converted to letter grades and grade points per the University calendar.

5. LEARNING OUTCOMES

1. Construct a technical project with real-world applications using sensors, actuators, computers or microprocessors with accompanied software logic and user interface.
2. Test the hardware and software components of the proposed design to evaluate its feasibility for implementation using the provided laboratory equipment and time line.
3. Construct a technical project with real-world applications using sensors, actuators, computers or microprocessors with accompanied software logic and user interface.
4. Understand the importance and necessity of team building and project management for complicated technical designs.
5. Generate technical reports and proposals to illustrate design plans using sophisticated and accurate technical language in appropriate report formats.
6. Deconstruct a technical problem into its constituent parts to systematically solve issues and incrementally move towards achieving an objective.
7. Judge the relevance of a design for its practicality and necessity in real-world applications.

6. COURSE OUTLINE – APPROVED ADVISORY STATEMENTS

ANTI-DISCRIMINATION

The Faculty of Engineering is concerned with ensuring an environment that is free of all discrimination. If there is a problem, individuals are reminded that they should contact the Department Chair, the Sexual Harassment Officer or the Human Rights Consultant, as soon as possible.
http://www.mcmaster.ca/policy/General/HR/Discrimination_Harassment_Sexual_Harassment-Prevention&Response.pdf

ACADEMIC INTEGRITY

You are expected to exhibit honesty and use ethical behaviour in all aspects of the learning process. Academic credentials you earn are rooted in principles of honesty and academic integrity. It is your responsibility to understand what constitutes academic dishonesty.

Academic dishonesty is to knowingly act or fail to act in a way that results or could result in unearned academic credit or advantage. This behaviour can result in serious consequences, e.g. the grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads: "Grade of F assigned for academic dishonesty"), and/or suspension or expulsion from the university. For information on the various types of academic dishonesty please refer to the Academic Integrity Policy, located at https://secretariat.mcmaster.ca/university-policies-procedures-guidelines/

The following illustrates only three forms of academic dishonesty: The following illustrates only three forms of academic dishonesty:

- plagiarism, e.g. the submission of work that is not one's own or for which other credit has been obtained.
- improper collaboration in group work.
- copying or using unauthorized aids in tests and examinations.

AUTHENTICITY / PLAGIARISM DETECTION

Some courses may use a web-based service (Turnitin.com) to reveal authenticity and ownership of student submitted work. For courses using such software, students will be expected to submit their work electronically either directly to Turnitin.com or via an online learning platform (e.g. A2L, etc.) using plagiarism detection (a service supported by Turnitin.com) so it can be checked for academic dishonesty.

Students who do not wish their work to be submitted through the plagiarism detection software must inform the Instructor before the assignment is due. No penalty will be assigned to a student who does not submit work to the plagiarism detection software. All submitted work is subject to normal verification that standards of academic integrity have been upheld (e.g., on-line search, other software, etc.). For more details about McMaster's use of Turnitin.com please go to www.mcmaster.ca/academicintegrity.

COURSES WITH AN ON-LINE ELEMENT

Some courses may use on-line elements (e.g. e-mail, Avenue to Learn (A2L), LearnLink, web pages, capa, Moodle, ThinkingCap, etc.). Students should be aware that, when they access the electronic components of a course using these elements, private information such as first and last names, user names for the McMaster e-mail accounts, and program affiliation may become apparent to all other students in the same course. The available information is dependent on the technology used. Continuation in a course that uses on-line elements will be deemed consent to this disclosure. If you have any questions or concerns about such disclosure please discuss this with the course instructor.

ONLINE PROCTORING

Some courses may use online proctoring software for tests and exams. This software may require students to turn on their video camera, present identification, monitor and record their computer activities, and/or lock/restrict their browser or other applications/software during tests or exams. This software may be required to be installed before the test/exam begins.

COMMUNICATIONS

It is the student's responsibility to:

- Maintain current contact information with the University, including address, phone numbers, and emergency contact information.
- Use the University provided e-mail address or maintain a valid forwarding e-mail address.
- Regularly check the official University communications channels. Official University communications are considered received if sent by postal mail, by fax, or by e-mail to the student's designated primary e-mail account via their @mcmaster.ca alias.
- Accept that forwarded e-mails may be lost and that e-mail is considered received if sent via the student's @mcmaster.ca alias.
- Check the McMaster/Avenue email and course websites on a regular basis during the term.

CONDUCT EXPECTATIONS

As a McMaster student, you have the right to experience, and the responsibility to demonstrate, respectful and dignified interactions within all of our living, learning and working communities. These expectations are described in the Code of Student Rights & Responsibilities (the "Code"). All students share the responsibility of maintaining a positive environment for the academic and personal growth of all McMaster community members, whether in person or online.

It is essential that students be mindful of their interactions online, as the Code remains in effect in virtual learning environments. The Code applies to any interactions that adversely affect, disrupt, or interfere with reasonable participation in University activities. Student disruptions or behaviours that interfere with university functions on online platforms (e.g. use of Avenue 2 Learn, WebEx or Zoom for delivery), will be taken very seriously and will be investigated. Outcomes may include restriction or removal of the involved students' access to these platforms.

ACADEMIC ACCOMMODATION OF STUDENTS WITH DISABILITIES

Students with disabilities who require academic accommodation must contact Student Accessibility Services (SAS) at 905-525-9140 ext. 28652 or sas@mcmaster.ca to make arrangements with a Program Coordinator. For further information, consult McMaster University's Academic Accommodation of Students with Disabilities policy.

REQUESTS FOR RELIEF FOR MISSED ACADEMIC TERM WORK

McMaster Student Absence Form (MSAF): In the event of an absence for medical or other reasons, students should review and follow the Academic Regulation in the Undergraduate Calendar "Requests for Relief for Missed Academic Term Work".

ACADEMIC ACCOMMODATION FOR RELIGIOUS, INDIGENOUS OR SPIRITUAL OBSERVANCES (RISO)

Students requiring academic accommodation based on religious, indigenous or spiritual observances should follow the procedures set out in the RISO policy. Students should submit their request to their Faculty Office normally within 10 working days of the beginning of term in which they anticipate a need for accommodation or to the Registrar's Office prior to their examinations. Students should also contact their instructors as soon as possible to make alternative arrangements for classes, assignments, and tests. http://www.mcmaster.ca/policy/Students-AcademicStudies/Studentcode.pdf

COPYRIGHT AND RECORDING

Students are advised that lectures, demonstrations, performances, and any other course material provided by an instructor include copyright protected works. The Copyright Act and copyright law protect every original literary, dramatic, musical and artistic work, including lectures by University instructors

The recording of lectures, tutorials, or other methods of instruction may occur during a course. Recording may be done by either the instructor for the purpose of authorized distribution, or by a student for the purpose of personal study. Students should be aware that their voice and/or image may be recorded by others during the class. Please speak with the instructor if this is a concern for you.

EXTREME CIRCUMSTANCES

The University reserves the right to change the dates and deadlines for any or all courses in extreme circumstances (e.g., severe weather, labour disruptions, etc.). Changes will be communicated through regular McMaster communication channels, such as McMaster Daily News, A2L and/or McMaster email.